

Electric Circuits Lab

Teacher : Behnaz Keshavarz Mohammadi
Student : Reyhan Sahraei, Ramtin Kosari

Review of Thevenin's Equivalent Circuit Theorem

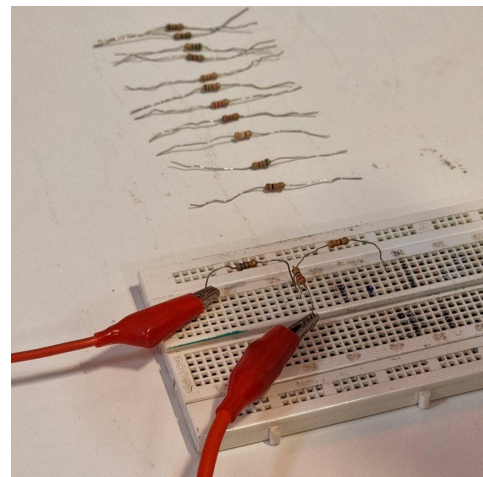
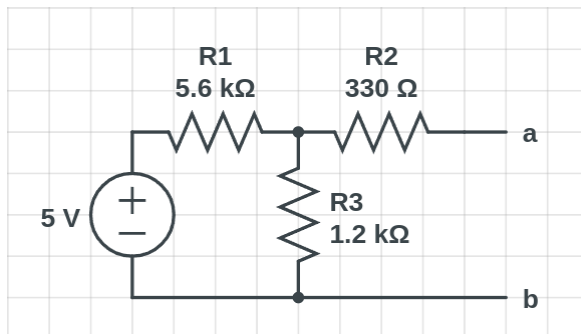
April 15, 2025

Overview

Here is book of Electric Circuits Lab experiment on page 65 (Book)

Experiment Info

- **Purpose** > To verify **Thevenin's Theorem** by determining the Thevenin equivalent voltage, current and resistance of a given electrical circuit.
- **Necessary Equipment** > Resistors, DC Power, Multimeter, Wires



Steps

1. First we set multimeter on voltage mode
2. Then we measured voltage from a and b and then measured current from there.
3. Then we removed the power source to make resistors form in parallel.

Thevenin Table (V-I)

$R_L (\Omega)$	Color 1	Color 2	Color 3	$V_L (V)$	$I_A (mA)$
$R_1 = 100\Omega$	Brown (1)	Black (0)	Brown (1)	0.061V	0.573mA
$R_2 = 150\Omega$	Brown (1)	Green (5)	Brown (1)	0.088V	0.590mA
$R_3 = 560\Omega$	Green (5)	Blue (6)	Brown (1)	0.257V	0.441mA
$R_4 = 820\Omega$	Gray (8)	Red (2)	Brown (1)	0.335V	0.388mA
$R_5 = 1.2k\Omega$	Brown (1)	Red (2)	Red (2)	0.415V	0.332mA
$R_6 = 1.5k\Omega$	Brown (1)	Green (5)	Red (2)	0.462V	0.298mA
$R_7 = 2.2k\Omega$	Red (2)	Red (2)	Red (2)	0.540V	0.243mA
$R_8 = 2.7k\Omega$	Red (2)	Violet (7)	Red (2)	0.584V	0.212mA
$R_9 = 3.9k\Omega$	Orange (3)	White (9)	Red (2)	0.650V	0.164mA
$R_{10} = 6.8\Omega$	Blue (6)	Gray (8)	Red (2)	0.730V	0.105mA

For a **Thevenin equivalent**, the V_L vs I_L curve should be **linear and decreasing** because :

$$V_L = V_{th} - I_L \cdot R_{th}$$

